

Lecture 8 - Multiparameter Models (Multivariate Gaussian)

DS 4420: Machine Learning and Data Mining 2

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Today

- 1 **Multiparameter Models**
- 2 **Multivariate Gaussians**
- 3 **MVN Marginalization**
- 4 **MVN Conditioning**
- 5 **Isotropic Posterior**

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- 2 Multivariate Gaussians
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- 5 Isotropic Posterior

Updates

- Homework 5 due **Monday, Mar. 23 by 11:59 pm**
 - Turn in by **Sunday, Mar. 22 at 11:59 pm** for 5% extra credit
 - Homework 6 (last HW) will be posted then, due **Monday, Apr. 6**
- Quiz 6 will be in-class **Thursday, Mar. 19** (on this material)
- TAs will host project check-in times the week of **Mar. 23**. Check-in at one of those sessions is **mandatory** for every group
- **Optional:** [Mid-Semester Course Eval Link](#) (or, QR-Code):



Topics for Today:

- 1 Multiparameter Models
- 2 Multivariate Gaussian
 - Marginalization
 - Conditioning
 - Posterior of Isotropic Gaussian

Push-Up Counter:

- Section 1: 13
- Section 4: 10

Today

1 Multiparameter Models

2 Multivariate Gaussians

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From Single to Multiparameter Models

Recall that in the previous lecture, we dealt with essentially only **single** parameter models, for example, the Beta-Bernoulli:

$$y|\theta \sim \text{Bern}(\theta)$$

$$\theta \sim \text{Beta}(\alpha, \beta)$$

The **Single** Parameter

Here the only parameter we assume we don't know is θ .

- θ completely characterizes the Bernoulli distribution of our data (which we will know once we collect it).
- The prior distribution for θ has two **hyper**parameters (which we are going to assume we know).

From Single to Multiparameter Models

There are several examples of distributions which have **multiple** parameters. A discrete example is the categorical:

$$y|\theta_1, \dots, \theta_K \sim \text{Cat}_K(\theta_1, \dots, \theta_K)$$

Notation and Priors

We often simplify the notation by calling $\theta = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_K \end{bmatrix} \in \mathbb{R}^K$

- We've also previously noted that the conjugate prior for the vector θ is the Dirichlet:

$$\theta \sim \text{Dirich}_K(\alpha)$$

- Where $\alpha^T = [\alpha_1 \quad \dots \quad \alpha_K] \in \mathbb{R}^K$
- In this case, while not trivial, we at least have a single (intuitive) distribution for modeling $\theta|y$

Dirichlet-Categorical Example (teased last lecture)

$$y|\theta \sim \text{Cat}_K(\theta), \quad y \in \{0, 1\}$$

$$\theta \sim \text{Dirich}_K(\alpha)$$

Goal: Find $p(\theta|y)$

Dirichlet-Categorical Example (teased last lecture)

Likelihood

$$p(y|\theta) = \prod_{i=1}^K \theta_i^{y_i}$$

Prior

$$p(\theta) = \frac{1}{B(\alpha)} \prod_{i=1}^K \theta_i^{\alpha_i - 1}$$

Joint

$$p(y, \theta) = \frac{1}{B(\alpha)} \prod_{i=1}^K \theta_i^{y_i + \alpha_i - 1}$$

Dirichlet-Categorical Example (teased last lecture)

Marginal

$$p(y) = \int_0^1 \int_0^1 \cdots \int_0^1 \frac{1}{B(\alpha)} \prod_{i=1}^K \theta_i^{y_i + \alpha_i - 1} d\theta_1 \cdots d\theta_{K-1} d\theta_K$$

Last lecture we saw how using the fact that $p(\theta|y) \propto p(y|\theta)p(\theta)$:

Posterior

$$p(\theta|y) \propto p(y, \theta) \propto \prod_{i=1}^K \theta_i^{y_i + \alpha_i - 1}$$

- Note that $\frac{1}{B(\alpha)}$ is a constant w.r.t θ , so we dropped it, which implies the posterior is:

$$\theta|y \sim \text{Dirich}_K(\alpha^*)$$

- where $\alpha_i^* = \alpha_i + y_i, i = 1, \dots, K$

Aside: Some Design Decisions

Note that the Dirichlet is a convenient choice because it is the conjugate prior, but it may be a **restrictive** choice as well. An [example](#) (from early in Dr. Gerber's PhD, where I would also call this an example of a **BAD** poster):

- In theory, the Categorical and Dirichlet distributions **both** assume the correlation between the K categories are all negative.
- What if this is not practical?
- You can use a different prior (for example, a [Logistic-Normal](#)) for θ which allows for positive correlations
- Unfortunately, it is **impossible** to marginalize the resulting joint distribution...
 - Problems like this motivate our use of [sampling techniques](#) (next lecture).

The Point

- Sometimes, the most convenient prior is **not** the best
- Of course, using a less convenient prior requires more work on our part to get the model to work...

Today

1 Multiparameter Models

2 Multivariate Gaussians

3 MVN Marginalization

4 MVN Conditioning

5 Isotropic Posterior

Multivariate Gaussians

The most common multiparameter distribution is without a doubt the Multivariate Gaussian, which we introduced in Lecture 6:

$$p(y) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(y-\mu)^T \Sigma^{-1} (y-\mu)}$$

Where:

- $y \in \mathbb{R}^d$
- $\mu \in \mathbb{R}^d$
- $\Sigma \in \mathbb{R}^{d \times d}$

Multivariate Gaussians

That's a lot of parameters...

Notice that not only do we now have $d + d + \frac{d^2-d}{2} = \frac{3d+d^2}{2}$ parameters, but these parameters represent fundamentally different aspects:

- There are d mean parameters making up the μ vector
- There are d variance and $(\frac{d^2-d}{2})$ covariance parameters making up the Σ matrix
- (Already quite a bit more complicated than the K probability parameters from the categorical)
- How do we determine appropriate prior distributions and then how are we going to marginalize to get the posterior?

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MVN Marginalization

We begin by discussing marginalization at the simplest level. The goal:

- Given the d variables that make up the $y \sim MVN_d(\mu, \Sigma)$ (for example, if $d = 5$):

$$p(y_1, y_2, \dots, y_5)$$

- We want to marginalize to get the joint distribution of some **subset** (for example):

$$p(y_1, y_3, y_5) = \int_{Y_4} \int_{Y_2} p(y) dy_2 dy_4$$

- Solving this integral is a very complicated process:

$$p(y_1, y_3, y_5) = \int_{Y_4} \int_{Y_2} p(y) dy_2 dy_4 = \int_{Y_4} \int_{Y_2} \frac{1}{(2\pi)^{5/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(y-\mu)^T \Sigma^{-1} (y-\mu)} dy_2 dy_4$$

MVN Marginalization

Luckily, mathematicians have already solved this for us, and (even more luckily) the solution is simple! Let's properly set everything up to understand the solution. We have:

$$p(y_1, y_3, y_5) = \int_{Y_4} \int_{Y_2} \frac{1}{(2\pi)^{5/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(y-\mu)^T \Sigma^{-1} (y-\mu)} dy_2 dy_4$$

where $y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix}$, $\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \end{bmatrix}$, and

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \sigma_{14} & \sigma_{15} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} & \sigma_{24} & \sigma_{25} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 & \sigma_{34} & \sigma_{35} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} & \sigma_4^4 & \sigma_{45} \\ \sigma_{51} & \sigma_{52} & \sigma_{53} & \sigma_{54} & \sigma_5^2 \end{bmatrix}$$

MVN Marginalization

First, given we want to keep y_1, y_3, y_5 , let's rearrange the definition of y and μ :

$$y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_3 \\ y_5 \\ y_2 \\ y_4 \end{bmatrix} = \begin{bmatrix} y_a \\ y_b \end{bmatrix} \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_3 \\ \mu_5 \\ \mu_2 \\ \mu_4 \end{bmatrix} = \begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}$$

MVN Marginalization

Correspondingly, we should rearrange Σ :

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{13} & \sigma_{15} & \sigma_{12} & \sigma_{14} \\ \sigma_{31} & \sigma_3^2 & \sigma_{35} & \sigma_{32} & \sigma_{34} \\ \sigma_{51} & \sigma_{53} & \sigma_5^2 & \sigma_{52} & \sigma_{54} \\ \sigma_{21} & \sigma_{23} & \sigma_{25} & \sigma_2^4 & \sigma_{24} \\ \sigma_{41} & \sigma_{43} & \sigma_{45} & \sigma_{42} & \sigma_4^2 \end{bmatrix} = \begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{bmatrix}$$

And, according to theory, we can actually state that the solution of the marginal $p(y_1, y_3, y_5) = p(y_a)$ is:

$$y_a \sim MVN_3(\mu_a, \Sigma_{aa})$$

- This solution tells us that the marginal distribution of a portion of a MVN is **also a MVN**:
 - With mean vector the vector of the means for the portion
 - And covariance matrix the variance-covariance terms for the portion
 - **Note:** the dimension of the conditional MVN is $d = 3$ because $y_a \in \mathbb{R}^3$
 - Depending on my mood, I may ask you to prove this on your homework...

Practice

A researcher is interested in the relationship between three variables in Boston: income level (X), internet access (Y), and education level (Z). These variables are modeled as a jointly Gaussian distribution:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \sim MVN_3(\mu = \begin{bmatrix} \mu_X \\ \mu_Y \\ \mu_Z \end{bmatrix} = \begin{bmatrix} 50 \\ 80 \\ 14 \end{bmatrix}, \Sigma = \begin{bmatrix} 100 & 10 & 20 \\ 10 & 30 & 15 \\ 20 & 15 & 25 \end{bmatrix})$$

The researcher wants to understand the relationship between X and Z independent of Y ; in other words, what is $p(x, z)$?

Context:

- X is measured in thousands of dollars per year
- Y is measured on a scale of 0 to 100, with higher scores indicating more internet access
- Z is measured in years of schooling (e.g. 12 years for high school)

Interpretation

Given the previous slides, it should be obvious that the distribution of X and Z (independent of Y) is:

$$\begin{bmatrix} X \\ Z \end{bmatrix} \sim MVN_2(\mu_{XZ} = \begin{bmatrix} \mu_X \\ \mu_Z \end{bmatrix} = \begin{bmatrix} 50 \\ 14 \end{bmatrix}, \Sigma_{XZ} = \begin{bmatrix} 100 & 20 \\ 20 & 25 \end{bmatrix})$$

Interpretation:

- Ignoring internet access gives us a more focused understanding of the relationship between income and education
- We note that the average income is 50 thousand dollars per year, and the average education level is 14 years of schooling, and that these two features are positively correlated

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MVN Conditioning

Now that we know how to marginalize a multivariate Gaussian, what about conditioning? What this means is, how do we find (for example):

$$p(y_2, y_4 | y_1, y_3, y_5)$$

How Would We Do This?

Well, by definition of conditional probability:

$$p(y_2, y_4 | y_1, y_3, y_5) = \frac{p(y_1, y_2, y_3, y_4, y_5)}{p(y_1, y_3, y_5)}$$

- Isn't that denominator something we just found?
- Still, the process of finding the conditional distribution is very... long.
- You are welcome to read about it on your own.
- We're going to simply jump to the conclusion.

MVN Conditioning

Borrowing everything from the MVN marginalization example, with:

$$y = \begin{bmatrix} y_1 \\ y_3 \\ y_5 \\ y_2 \\ y_4 \end{bmatrix} = \begin{bmatrix} y_a \\ y_b \end{bmatrix}, \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_3 \\ \mu_5 \\ \mu_2 \\ \mu_4 \end{bmatrix} = \begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{bmatrix}$$

Our **goal** is to find $p(y_b | y_a)$. It can be shown that this is simply also a Gaussian:

$$y_b | y_a \sim MVN_2(\mu_{b|a}, \Sigma_{b|a})$$

Where:

- $\mu_{b|a} = \mu_b + \Sigma_{ba} \Sigma_{aa}^{-1} (y_a - \mu_a)$
 - **Note:** the dimension of the conditional MVN is $d = 2$ because $y_b \in \mathbb{R}^2$
- $\Sigma_{b|a} = \Sigma_{bb} - \Sigma_{ba} \Sigma_{aa}^{-1} \Sigma_{ab}$

Practice In Theory

Given the same 5-dimensional Gaussian as before, find the conditional distribution of $p(\mathbf{y}_a | \mathbf{y}_b)$.

Practice With Data

Professor Pine is a Pokémon professor and is studying how health (y_1 , from 1 to 255), height (y_2 , in cm), weight (y_3 , in kg), and speed (y_4 , in m/s) of a Pokémon are all related. She models these four variables as a Gaussian distribution:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \sim MVN_4(\mu = \begin{bmatrix} 64 \\ 47 \\ 101 \\ 69 \end{bmatrix}, \Sigma = \begin{bmatrix} 800 & 260 & 1670 & -30 \\ 260 & 1430 & 2810 & 200 \\ 1670 & 2810 & 17240 & 190 \\ -30 & 200 & 190 & 730 \end{bmatrix})$$

- Dr. Gerber's favorite Pokémon is Cubone:

$$y_{cubone} = \begin{bmatrix} 50 \\ 16 \\ 14 \\ 35 \end{bmatrix}$$

Practice With Data

Given the information on the previous page:

- ① Plot out the distribution of Cubone's health given his other features: $p(y_1|y_2, y_3, y_4)$ and draw a vertical line around his actual health. Do his features correspond to his actual health, or would you expect him to be healthier/unhealthier?
- ② Is weight dependent on health? Identify $p(y_1, y_3)$ and calculate the correlation between health and weight to quantify their dependency.
- ③ Plot $p(y_2, y_3|y_1, y_4)$ in 3D, and place a point at Cubone's location. How does his height and weight compare to the population of Pokémon in his health and speed range?
- ④ What insights might you use, as a Pokémon professor, to improve Cubone's health?

Solution 1

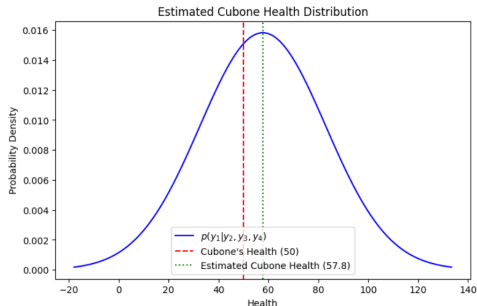
We want to find $p(y_1|y_2, y_3, y_4)$ and to plot it. This will be a **univariate** Gaussian. Given:

$$y_a = y_1, \quad y_b = \begin{bmatrix} y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 16 \\ 14 \\ 35 \end{bmatrix}$$

- $\mu_a = 64, \mu_b = \begin{bmatrix} 47 \\ 101 \\ 69 \end{bmatrix}$
- $\Sigma_{aa} = 800$
- $\Sigma_{bb} = \begin{bmatrix} 1430 & 2810 & 200 \\ 2810 & 17240 & 190 \\ 200 & 190 & 730 \end{bmatrix}$
- $\Sigma_{ab} = \begin{bmatrix} 260 & 1670 & -30 \end{bmatrix}$
- $\mu_{a|b} = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(y_b - \mu_b)$
 - $64 + \Sigma_{ab}\Sigma_{bb}^{-1} \left(\begin{bmatrix} 16 \\ 14 \\ 35 \end{bmatrix} - \begin{bmatrix} 47 \\ 101 \\ 69 \end{bmatrix} \right)$
 - $\mu_{a|b} = 57.8$
- $\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$
 - $800 - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$
 - $\Sigma_{a|b} = 635$

You can see all this work done in Python and R, and create the plot on the next page, in the **mvnmodels** files.

Solution 1



Note, that we would say:

$$y_1|y_2, y_3, y_4 \sim N(\mu_{1| \cdot} = 57.8, \sigma_{1| \cdot}^2 = 635)$$

- Where this is the distribution specifically for Cubone's health (given his height, weight, and speed).
- And we use the $1| \cdot$ notation to indicate it is the distribution of y_1 given everything else.

Solution 2

Now we are interested in $p(y_1, y_3)$; the unconditional joint distribution of a Pokémon's health and weight. We know from theory this is simple:

$$\begin{bmatrix} y_1 \\ y_3 \end{bmatrix} \sim MVN_2(\mu_{13} = \begin{bmatrix} 64 \\ 101 \end{bmatrix}, \Sigma_{13} = \begin{bmatrix} 800 & 1670 \\ 1670 & 17240 \end{bmatrix})$$

- From here, the correlation can be calculated easily using Σ_{13} :

$$\rho_{13} = \frac{1670}{\sqrt{800}\sqrt{17240}} = 0.45$$

- A moderate positive relationship.

Solution 3

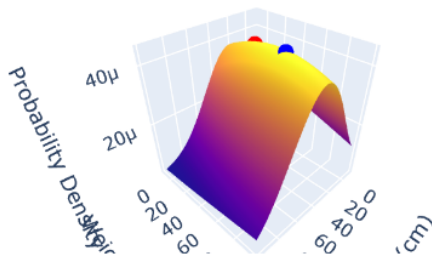
Finally, we are interested in $p(y_2, y_3 | y_1, y_4)$ and in plotting this **bivariate** Gaussian in 3D with a reference point for Cubone. The distribution can be found given:

$$y_a = \begin{bmatrix} y_2 \\ y_3 \end{bmatrix}, \quad y_b = \begin{bmatrix} y_1 \\ y_4 \end{bmatrix} = \begin{bmatrix} 50 \\ 35 \end{bmatrix}$$

- $\mu_a = \begin{bmatrix} 47 \\ 101 \end{bmatrix}, \mu_b = \begin{bmatrix} 64 \\ 69 \end{bmatrix}$
- $\Sigma_{aa} = \begin{bmatrix} 1430 & 2810 \\ 2810 & 17240 \end{bmatrix}$
- $\Sigma_{bb} = \begin{bmatrix} 800 & -30 \\ -30 & 730 \end{bmatrix}$
- $\Sigma_{ab} = \begin{bmatrix} 260 & 200 \\ 1670 & 190 \end{bmatrix} \Rightarrow \Sigma_{ba} = \Sigma_{ab}^T = \begin{bmatrix} 260 & 1670 \\ 200 & 190 \end{bmatrix}$
- $\mu_{a|b} = \begin{bmatrix} 32.5 \\ 59.8 \end{bmatrix}$
- $\Sigma_{a|b} = \begin{bmatrix} 1285 & 2195 \\ 2195 & 13666 \end{bmatrix}$

You can see all this work done in Python and R, and create the plot on the next page, in the **mvnmodels** files.

Solution 3



Note, that we would say:

$$y_2, y_3 | y_1, y_4 \sim MVN_2(\mu_{a|b} = \begin{bmatrix} 32.5 \\ 59.8 \end{bmatrix}, \Sigma_{a|b}^2 = \begin{bmatrix} 1285 & 2195 \\ 2195 & 13666 \end{bmatrix})$$

- Where this is the distribution specifically for Cubone's height and weight (given his health and speed).

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Normal Bayesians (or, Bayesian Gaussians?)

So, given a general multivariate Gaussian, $p(y_1, y_2, \dots, y_d)$ we are able to find both marginal and conditional distributions (luckily, all Gaussian).

- But what about moving to a Bayesian modeling context?
- On Homework 5, you are told that the conjugate prior for a Gaussian mean μ is also Gaussian such that:

The Gaussian is a prior for The Gaussian (sometimes)

$$y|\mu, \sigma \sim N(\mu, \sigma = \sigma_0)$$

$$\mu \sim N(\eta, \tau)$$

$$\mu|y, \sigma \sim N(\mu^*, \sigma^*)$$

Where:

- $\mu^* = \frac{\frac{n\bar{y}}{\sigma^2} + \frac{\eta}{\tau^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$
- $\sigma^* = \sqrt{\left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1}}$
- **But NOTE:** this is for a **univariate** Gaussian and assumes that we **know** the standard deviation $\sigma = \sigma_0$
 - What if we have a multivariate Gaussian? What if we don't know Σ ?

The Normal-Inverse-Gamma Prior for the Isotropic Gaussian

For this complicated situation, let's use the simplest of the multivariate Gaussian. Recall the **Isotropic** Gaussian:

$$p(y|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{d/2}} e^{-\frac{1}{2\sigma^2} \|y-\mu\|_2^2}$$

- The **conjugate** prior for the mean and variance is the **Normal-Inverse-Gamma**, such that:

$$p(\mu, \sigma^2) = p(\mu|\sigma^2)p(\sigma^2)$$

- $\mu|\sigma^2 \sim N(\mu_0, \sigma^2 \lambda_0^{-1})$
- $\sigma^2 \sim \text{Gamma}^{-1}(\alpha_0, \beta_0)$

The Normal-Inverse-Gamma Prior for the Isotropic Gaussian

Note this joint prior distribution is quite complicated looking, since it is the product of the two distributions $p(\mu|\sigma^2)$ and $p(\sigma^2)$, where:

$$p(\mu|\sigma^2) = \frac{1}{(2\pi \frac{\sigma^2}{\lambda_0})^{d/2}} e^{-\frac{\lambda_0}{2\sigma^2} \|\mu - \mu_0\|_2^2}$$

$$p(\sigma^2) = \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)} (\sigma^2)^{-\alpha_0-1} e^{-\frac{\beta_0}{\sigma^2}}$$

Once You Know The Prior Is Conjugate...

To help us figure out the posterior distribution, $\mu, \sigma^2|y$, let's simplify the problem a bit.

- Once you know the prior is conjugate, you know that all you need to do is **update the parameters** of the prior given the data
- Note that for the very first Bayesian model we learned, the Beta-Bernoulli:

$$y|\theta \sim \text{Bern}(\theta)$$

$$\theta \sim \text{Beta}(\alpha_0, \beta_0)$$

So:

$$p(\theta) = \frac{\theta^{\alpha_0-1}(1-\theta)^{\beta_0-1}}{B(\alpha_0, \beta_0)}$$

And that the posterior could be written as:

$$p(\theta|y) = \frac{\theta^{\alpha_1-1}(1-\theta)^{\beta_1-1}}{B(\alpha_1, \beta_1)}$$

Where $\alpha_1 = \alpha_0 + y$ and $\beta_1 = \beta_0 + 1 - y$

Once You Know The Prior Is Conjugate...

We can then say the same thing in this more complicated situation: since the Normal-Inverse-Gamma is the conjugate prior for the Isotropic Gaussian, the posterior must also be Normal-Inverse-Gamma:

$$p(\mu, \sigma^2) = \frac{1}{(2\pi \frac{\sigma^2}{\lambda_0})^{d/2}} e^{-\frac{\lambda_0}{2\sigma^2} \|\mu - \mu_0\|_2^2} \times \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)} (\sigma^2)^{-\alpha_0-1} e^{-\frac{\beta_0}{\sigma^2}}$$

Or, $\mu, \sigma^2 \sim NG^{-1}(\mu_0, \lambda_0, \alpha_0, \beta_0)$. The posterior must be some form of:

$$p(\mu, \sigma^2 | y) = \frac{1}{(2\pi \frac{\sigma^2}{\lambda_1})^{d/2}} e^{-\frac{\lambda_1}{2\sigma^2} \|\mu - \mu_1\|_2^2} \times \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)} (\sigma^2)^{-\alpha_1-1} e^{-\frac{\beta_1}{\sigma^2}}$$

Or, $\mu, \sigma^2 | y \sim NG^{-1}(\mu_1, \lambda_1, \alpha_1, \beta_1)$, where each of those parameters is updated and/or depends on y (the data) somehow. In fact, the solution (which I may or may not ask you to do on a homework...) is:

$$\mu_1 = \frac{y + \lambda_0 \mu_0}{1 + \lambda_0}, \quad \lambda_1 = 1 + \lambda_0, \quad \alpha_1 = \frac{d}{2} + \alpha_0, \quad \beta_1 = \frac{1}{2} (y - \mu_0)^T (y - \mu_0) \frac{\lambda_0}{1 + \lambda_0} + \beta_0$$

Practice

You work for a self-driving car company and your job is to work with the tracking of detected objects. Here is the set-up:

- The sensors detect objects every 1 millisecond and treats the current car location as $\hat{y} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- Coordinates are such that $\begin{bmatrix} 0 & 1 \end{bmatrix}^T$ is 1 meter in front of the car and $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$ is 1 meter to the right of the car (so, $\begin{bmatrix} -1 & 0 \end{bmatrix}^T$ is one meter to the left)
- The standard deviation of each sensor (forward and lateral) is $\sigma = 0.5$
- Given that you detect an object initially at $\begin{bmatrix} 0 & 3 \end{bmatrix}^T$, if the next measurement is at $\begin{bmatrix} 0.1 & 2.8 \end{bmatrix}^T$, what is the expected location of the object?
- If we receive the same readings ($\begin{bmatrix} 0.1 & 2.8 \end{bmatrix}^T$) for the next 3 measurements, how does the likely (posterior) location (μ) **and uncertainty** (σ) of the object change? What do these changes mean?

The instructions for answering these questions are provided on the next slide, followed by the solutions. **Please don't look at the solutions until you've tried it yourself.**

Practice Instructions

- 1 Use $\mu_0 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$ and $\lambda_0 = 1$
- 2 Since the sensors $\sigma = .5 \Rightarrow \sigma^2 = .25$, we want the $E(\sigma^2) = .25 = \frac{\beta_0}{\alpha_0 - 1}$ to choose the appropriate α_0 and β_0 :
 - While there are many values we could use, let's use $\alpha_0 = 5$ and $\beta_0 = 1$
- 3 Once we have observed $y = \begin{bmatrix} 0.1 \\ 2.8 \end{bmatrix}$, we can use it to evaluate the posterior (from slide 38)
- 4 After each iteration (new sample) we use the previously updated values as the “prior” values to perform the update.

Practice Solutions

In the instructions, you were given:

$$\mu_0 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \quad \lambda_0 = 1, \quad \alpha_0 = 5, \quad \beta_0 = 1, \quad y = \begin{bmatrix} 0.1 \\ 2.8 \end{bmatrix}$$

So, the posterior parameters (from slide 38) are:

$$\mu_1 = \frac{y + \lambda_0 \mu_0}{1 + \lambda_0}, \quad \lambda_1 = 1 + \lambda_0, \quad \alpha_1 = \frac{d}{2} + \alpha_0, \quad \beta_1 = \frac{1}{2} (y - \mu_0)^T (y - \mu_0) \frac{\lambda_0}{1 + \lambda_0} + \beta_0$$

$$\mu_1 = \begin{bmatrix} 0.05 \\ 2.9 \end{bmatrix}, \quad \lambda_1 = 2, \quad \alpha_1 = 6, \quad \beta_1 = 1.0125$$

- The posterior mean of μ is μ_1 (our updated belief about where the object is)
- The posterior mean of σ is $\sqrt{\frac{\beta_1}{\alpha_1 - 1}} = 0.45$

Practice Solutions

For the next three readings, we simply do everything again, given:

$$\mu_1 = \begin{bmatrix} 0.05 \\ 2.9 \end{bmatrix}, \quad \lambda_1 = 2, \quad \alpha_1 = 6, \quad \beta_1 = 1.0125, \quad y = \begin{bmatrix} 0.1 \\ 2.8 \end{bmatrix}$$

The posterior parameters (from slide 38) are:

$$\mu_2 = \begin{bmatrix} 0.067 \\ 2.867 \end{bmatrix}, \quad \lambda_2 = 3, \quad \alpha_2 = 7, \quad \beta_2 = 1.0167$$

- The posterior mean of μ is μ_2 (our updated belief about where the object is)
- The posterior mean of σ is $\sqrt{\frac{\beta_2}{\alpha_2 - 1}} = 0.412$

See if you can complete the next two measurements on your own and finally answer the important question: as we collect more observations (that are all the same), what is happening to our uncertainty about where the object is?